

Harikeshav Rameshkumar

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EDUCATION

The Ohio State University

B.S. Computer Science | GPA: 3.9/4.0

May 2027
Columbus, OH

- **Courses:** Computer Organization, Data Structures & Algorithms, Foundations of Software Engineering
- **Honors:** Dean's List, University Honors

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Rust, TypeScript
Technologies: LangChain/LangGraph, PyTorch, FastAPI, React
DevOps: Docker, Linux, GitHub Actions, AWS, PostgreSQL

EXPERIENCE

Apprenticeship

Jan. 2026 – Present

Pfizer

Remote

- **Architected an end-to-end document automation platform** for pharmaceutical and mortgage sectors, utilizing **Python, PyMuPDF, and OCR engines (Tesseract, PaddleOCR)** to extract and classify data from massive unstructured "blobs" with custom regex and layout-based heuristics.
- **Developed a high-precision RAG pipeline using LlamaIndex and Gradio**, implementing advanced chunking and metadata filtering to optimize retrieval across open-source models like **Mistral and Phi-2**, resulting in a modular system capable of real-time querying and automated compliance flagging.

Software Engineering Internship

Jun. 2025 – Aug. 2025

Siage Solutions

Bangalore, IN

- **Engineered a scalable, secure RAG architecture and ML ticketing system** that indexed **100+ technical manuals** and automated prioritization for **10+ enterprise clients**; utilized vector databases and semantic embeddings to eliminate ticket redundancy and reduce documentation search time.
- **Developed an end-to-end data intelligence pipeline** to scrape and analyze **50,000+ multi-regional product reviews**, leveraging LLMs for sentiment analysis and thematic extraction to deliver actionable insights via automated trend visualizations and heatmaps.

Research Internship

Jun. 2023 – Oct. 2023

Indian Institute of Technology, Madras

Remote

- **Engineered a predictive ML framework** to model indoor wireless signal propagation using electromagnetic wave theory; conducted large-scale experiments to train models capable of forecasting signal strength across diverse architectural geometries.
- **Authored a comprehensive report** evaluating comparative model performance and generating high-resolution spatial heat maps to optimize indoor network planning and deployment strategies.

PROJECTS

Pulse | Python, FastAPI, LangGraph, React Native, Redis

2026

- Developed an **AI-powered clinical trial monitoring platform** leveraging **LangGraph and LiveKit** to facilitate real-time conversational symptom reporting, automatically mapping free-text to MedDRA clinical coding standards.
- Engineered an **event-driven anomaly detection pipeline** utilizing **FastAPI, Redis, and WebSockets** to passively ingest wearable health metrics, calculating composite risk scores and broadcasting alerts to a **React** dashboard.

LEAP | C++20, Linux Kernel, LibTorch

2026

- Engineered a **distributed LLM inference engine in C++20** that pipelines **70B+ parameter models** across consumer devices via a **Ring Topology**, utilizing **INT8 quantization** to solve VRAM bottlenecks.
- Implemented a custom **Linux Kernel Module** for **zero-copy networking** and hand-written **AVX2/NEON SIMD kernels**, achieving **near-native throughput** and enabling real-time inference on heterogeneous hardware.

Distill | Python, PyTorch, BERT, React

2026

- Developed an **intelligent LLM context compression library** using a fine-tuned **BERT token classification model** to prune semantic noise, reducing prompt size by **68%** while retaining **99% accuracy** on LongBench V2.
- Built a two-tier filtering pipeline with **word-level aggregation** and **force-token preservation** to prevent hallucinations, achieving **37% lower latency** and **68% cost reduction** for GPT-4o inference.

LeadForge | Python, LangGraph, React, Docker

2025

- Architected an **autonomous AI SDR platform** using **LangGraph and Gemini** that orchestrates lead generation and multi-channel outreach, automating the pipeline from Google Maps scouting to **Twilio** voice calls.
- Developed a **generative microservices workflow** that autonomously critiques prospect websites using visual AI and code-generates improved **React/Tailwind prototypes**, deploying them instantly to drive conversion.

Sane Jtreet | Python, LangGraph, FastAPI, ChromaDB

2025

- Engineered a **multi-agent trading platform** where specialized LLM agents (Analysts, Debaters, Risk Managers) autonomously collaborate via **LangGraph** to analyze real-time market/sentiment data and formulate investment strategies.
- Implemented a **cognitive architecture with memory reflection** using **ChromaDB**, enabling agents to store past trade outcomes and retrieve "lessons learned" to iteratively improve predictive accuracy and risk management.

AWARDS

Second Place - Medpace Track | RevolutionUC

2026

- Awarded for **Pulse**, an AI-powered clinical trial safety platform integrating conversational symptom journaling, wearable health monitoring, and real-time researcher dashboards.

Third Place - Visa Track | TartanHacks

2026

- Awarded for **Penny**, a gamified AI-driven personal finance assistant that simplifies expense tracking and financial goal setting through intelligent receipt analysis and interactive mascot-led insights.

Third Place - The Token Company Track | NextHacks

2026

- Awarded for **Distill**, a high-performance LLM context compression library enabling drastic reductions in inference costs and latency while preserving accuracy across massive context windows.

Best AI Hack Runner Up | HackOHIO

2025

- Awarded for **LeadForge**, an autonomous AI SDR platform that orchestrates lead generation and multi-channel outreach using LangGraph and Gemini.

Game Development | World Language Appathon

2025

- Built an immersive VR market simulator to teach the player Uzbek, featuring a dynamic economy system, physics-based interactions, and AI-driven NPCs, developed for Meta Quest using Unity and the Meta XR SDK.

Various Wins | Regional Hackathons

2022 – 2024

- Participated in multiple regional hackathons and won multiple prizes.